



Industrial Consulting Project

MVP Development of a Secure Multi-Vendor Product Discovery Platform

1. Project Overview

This project requires students to design and build a working Minimum Viable Product (MVP) for a secure online platform that allows products from multiple vendors to be listed, searched, filtered, compared, and viewed by users.

The focus is not only technical development, but also strong user experience design. Students must carry out user research, create personas, develop wireframes, map user journeys, produce mockups, and build a usable, secure, responsive web application.

The final MVP should demonstrate a polished front-end interface, a functional backend, secure user handling, and a database-driven product listing system.

2. Project Aim

To design and develop a secure, user-friendly multi-vendor product discovery platform that enables users to browse, search, filter, and view products from different vendors in one place.

3. Suggested Technologies

Students should use the following:

Frontend:

- React
- JavaScript or TypeScript
- HTML
- CSS
- Tailwind CSS or Bootstrap, optional

Backend:

- Node.js
- Express.js

Database options:



- SQL, such as SQLite, MySQL or PostgreSQL
- Firebase
- MongoDB

Other tools:

- GitHub
- Figma
- Canva or Adobe XD
- Postman
- Trello, Jira or GitHub Projects

Phase 1: Research, UX Design and Planning

Weeks 1–4

1.1 Understand the Problem

Students should investigate how users currently search for products across different websites.

Tasks:

1. Identify common problems users face when browsing products online.
2. Explore how users compare prices, brands, reviews and features.
3. Investigate what makes a product discovery website trustworthy.
4. Review 3–5 existing product listing or marketplace websites.
5. Identify good and poor design patterns.

Deliverables:

- Competitor analysis
- Problem statement
- Initial feature list

1.2 User Research

Students must conduct basic user research.

Tasks:

1. Interview at least 10 potential users.
2. Ask users how they search for products online.



3. Ask what frustrates them when comparing products.
4. Ask what filters and information they expect.
5. Summarise findings into key themes.

Example interview questions:

- How do you usually search for products online?
- What makes you trust a product website?
- What information do you need before clicking on a product?
- What filters do you normally use?
- What would make you leave a website quickly?

Deliverables:

- Interview questions
- Interview summary
- Key user needs
- Evidence of research

1.3 Personas

Create at least 3 user personas.

Each persona should include:

- Name
- Age range
- Goals
- Frustrations
- Technical confidence
- Buying behaviour
- Accessibility needs, if relevant

Example personas:

1. Busy professional looking for quick comparisons
2. Budget-conscious shopper



3. Quality-focused shopper looking for trusted brands

Deliverables:

- 3 detailed personas

1.4 User Journey Mapping

Students must map the journey from landing on the website to viewing or clicking a product.

Minimum user journey:

1. User visits homepage
2. User searches or chooses a category
3. User applies filters
4. User views product cards
5. User opens product details
6. User clicks external vendor link

Deliverables:

- User journey map
- Pain points
- Opportunities for improvement

1.5 Information Architecture

Students should design the structure of the platform.

Required pages:

- Homepage
- Product listing page
- Product details page
- Vendor listing page
- Login page
- Register page
- Admin dashboard
- Add product page



- Edit product page
- About page
- Privacy page

Deliverables:

- Sitemap
- Navigation structure

1.6 Wireframes

Create low-fidelity wireframes first.

Required wireframes:

- Homepage
- Product listing page
- Product details page
- Login/register page
- Admin dashboard
- Add product form

Deliverables:

- Low-fidelity wireframes
- Explanation of design decisions

1.7 High-Fidelity Mockups

Students should convert selected wireframes into polished mockups.

The design should include:

- Clear branding
- Consistent colour palette
- Clear typography
- Product cards
- Filter sidebar
- Search bar
- Responsive layout



- Mobile-friendly screens

Deliverables:

- High-fidelity mockups
- Mobile and desktop versions

Phase 2: MVP Development

Weeks 5–8

2.1 Project Setup

Tasks:

1. Create GitHub repository.
2. Set up React frontend.
3. Set up Express backend.
4. Set up database.
5. Create project folder structure.
6. Create development branches.

2.2 Database Design

The database should support products, vendors and users.

Minimum tables or collections:

Users

- user_id
- name
- email
- password_hash
- role
- created_at

Vendors

- vendor_id
- vendor_name
- website_url



- logo_url
- description

Products

- product_id
- vendor_id
- title
- description
- category
- price
- image_url
- product_url
- availability
- created_at

Categories

- category_id
- category_name

Deliverables:

- Database schema
- Entity relationship diagram
- Seed data with at least 30 products
- At least 5 vendors

2.3 Frontend Features

Students must build a responsive interface.

Required frontend features:

1. Homepage with hero section
2. Search bar
3. Product categories
4. Product listing grid



5. Product cards
6. Product detail page
7. Filters by category, price and vendor
8. Sort by price or newest
9. Vendor profile section
10. Responsive mobile layout

Product card should show:

- Product image
- Product title
- Vendor name
- Price
- Category
- View details button

Deliverables:

- React components
- Responsive pages
- Clean CSS styling

2.4 Backend Features

Required API routes:

GET /api/products

GET /api/products/:id

GET /api/vendors

GET /api/vendors/:id

GET /api/categories

POST /api/auth/register

POST /api/auth/login

POST /api/products

PUT /api/products/:id



DELETE /api/products/:id

Admin-only routes:

POST /api/products

PUT /api/products/:id

DELETE /api/products/:id

Deliverables:

- Express server
- REST API
- Tested endpoints
- Postman screenshots or API documentation

2.5 Authentication and Authorisation

The platform must include secure login.

Required features:

1. User registration
2. User login
3. Password hashing
4. Session or JWT-based authentication
5. Role-based access control
6. Admin-only product management

Roles:

- Visitor: can browse products
- Registered user: can save or favourite products, optional
- Admin: can add, edit and delete products

Deliverables:

- Secure authentication
- Admin-protected pages
- Evidence of testing

2.6 Product Management Dashboard



Admins should be able to manage products.

Required dashboard features:

1. View all products
2. Add new product
3. Edit product
4. Delete product
5. Assign product to vendor
6. Assign category
7. Add image URL
8. Add external product link

Deliverables:

- Admin dashboard
- Product management forms
- CRUD functionality

Phase 3: Testing, Security, Evaluation and Final Delivery

Weeks 9–12

3.1 Usability Testing

Students must test the platform with at least 5 users.

Example tasks:

1. Find a product using search.
2. Filter products by category.
3. View product details.
4. Click through to vendor website.
5. Login as admin and add a product.

Students should record:

- Task success rate
- Time taken
- User comments



- Problems observed
- Suggested improvements

Deliverables:

- Usability testing plan
- Testing results
- Improvements made

3.2 Security Testing

Students must demonstrate basic security awareness.

Required checks:

1. Passwords are hashed.
2. Admin pages are protected.
3. Users cannot delete products without admin role.
4. Forms validate input.
5. API routes validate data.
6. SQL injection or NoSQL injection risks are considered.
7. Sensitive keys are not stored publicly.
8. Error messages do not expose system details.

Deliverables:

- Security checklist
- Evidence of fixes
- Short security reflection

3.3 Functional Testing

Students should test all key features.

Minimum test cases:

Feature	Expected Result
Search product	Matching products displayed
Filter by category	Only selected category shown



Filter by vendor	Only selected vendor shown
View details	Correct product details shown
Admin login	Admin dashboard accessible
Add product	Product saved to database
Edit product	Product details updated
Delete product	Product removed
Invalid login	Error shown
Protected route	Non-admin blocked

Deliverables:

- Test table
- Screenshots
- Bug fixing log

3.4 Accessibility and Responsiveness

The platform should be usable on desktop, tablet and mobile.

Students should check:

1. Text contrast
2. Keyboard navigation
3. Clear buttons
4. Alt text for images
5. Mobile layout
6. Form labels
7. Error messages
8. Readable font sizes

Deliverables:

- Accessibility checklist
- Mobile screenshots
- Desktop screenshots



3.5 Final MVP Requirements

The final MVP must include:

1. Working homepage
2. Product listing page
3. Product detail page
4. Vendor data
5. Product search
6. Product filtering
7. Admin login
8. Add/edit/delete products
9. Secure authentication
10. Responsive design
11. Database connection
12. Seed product data
13. GitHub repository
14. README file
15. Final demo video

Presentation: Adhering to the university guidelines, the presentation should incorporate a live demonstration of both the backend and front-end components of the project.

Contact Details:

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Resource Requirements: To be discussed as and when needed.

Note to the client: Please refer to the attached client guidance document with sample of projects for more clarity.